$$\begin{cases} X_{24} = X_{14} - X_{13} \\ X_{2j} = X_{1j} - X_{13} & j = 5, \dots, n-2 \\ X_{2n-1} = X_{1,n-1} - X_{13} \\ X_{2n} = -X_{13} \end{cases}$$

starts highlighted

+0 proj: $\rightarrow$

$$X_{2j} = X_{1j} - X_{13} \quad j = 4, \dots, n-2$$
$$X_{2n-1} = -X_{13}$$

+0

−1 proj:

$$X_{13} = X_{n3} - X_{n2}$$
$$X_{1j} = X_{nj} - X_{n2} \quad j = 4, \dots, n-3$$
$$X_{1n-2} = X_{nn-2} - X_{n2}$$
$$X_{1n-1} = -X_{n2}$$

$\xrightarrow{X_{1,n-1}=0}$

$$X_{13} = X_{n3}$$
$$X_{1j} = X_{nj}$$
$$X_{1n-2} = X_{nn-2}$$
$$X_{n2} = 0$$

×

(0 - zero, i.e. no contribution on $1, \dots, n-1$)

−2 proj:

$$X_{n2} = X_{n-12} - X_{n-11}$$
$$X_{nj} = X_{n-1j} - X_{n-1,1} \quad (j = 3, \dots, n-4)$$
$$X_{nn-3} = X_{n-1n-3} - X_{n-1,1}$$
$$X_{n,n-2} = -X_{n-1,1}$$

$\xrightarrow{X_{1n-1}=0}$

$$X_{n2} = X_{n-12}$$
$$X_{nj} = X_{n-1j}$$
$$X_{nn-3} = X_{n-1n-3}$$
$$X_{n,n-2} = 0$$

×

(0 - zero, i.e. no contribution on $1, \dots, n-1$)

−3 proj:

$$X_{n-1,1} = X_{n-2,1} - X_{n-2,n}$$
$$X_{n-1,j} = X_{n-2,j} - X_{n-2,n} \quad (j = 2, \dots, n-5)$$
$$X_{n-1,n-4} = X_{n-2,n-4} - X_{n-2,n}$$
$$X_{n-1,n-3} = -X_{n-2,n}$$

$\xrightarrow{X_{1,n-1}=0}$

$-2 \equiv +(n-1-2)$

$$X_{n-1,j} = X_{n-2,j} - X_{n-2,1} \quad j = (2, n-4)$$
$$X_{n-1,n-3} = -X_{n-2,1}$$

+1 proj:

$$X_{3j} = X_{2j} - X_{24} \quad (j = 5, \dots, n)$$
$$X_{31} = -X_{24}$$

$\rightarrow$

+1

$$X_{3j} = X_{2j} - X_{24} \quad (j = 5, \dots, n-1)$$
$$X_{31} = -X_{24}$$

+2 proj:

$$X_{4j} = X_{3j} - X_{35} \quad (j = 6, \dots, n)$$
$$X_{41} = X_{31} - X_{35}$$
$$X_{42} = -X_{35}$$

$\rightarrow$

+2

$$X_{4j} = X_{3j} - X_{35} \quad (j = 7, \dots, n-1)$$
$$X_{41} = X_{31} - X_{35}$$
$$X_{42} = -X_{35}$$

+3 proj:

$$X_{5j} = X_{4j} - X_{46} \quad (j = 7, \dots, n)$$
$$X_{5j} = X_{4j} - X_{46} \quad (j = 1, 2)$$
$$X_{53} = -X_{46}$$

$\rightarrow$

$$X_{5j} = X_{4j} - X_{46} \quad (j = 7, \dots, n-1)$$
$$X_{5j} = X_{4j} - X_{46} \quad (j = 1, 2)$$
$$X_{53} = -X_{46}$$

+3

in general, (+ℓ) with $\ell \in \{1, \dots, n-4\}$

$$X_{2+\ell,j} = X_{1+\ell,j} - X_{1+\ell,3+\ell} \qquad j \in \{4+\ell, \dots, n\}$$
$$X_{2+\ell,j} = X_{1+\ell,j} - X_{1+\ell,3+\ell} \qquad j \in \{1, \ell-1\}$$
$$X_{2+\ell,\ell} = -X_{1+\ell,3+\ell}$$

$\rightarrow$

$$X_{2+\ell,j} = X_{1+\ell,j} - X_{1+\ell,3+\ell} \quad (j = 4+\ell, \dots, n-1)$$
$$X_{2+\ell,j} = X_{1+\ell,j} - X_{1+\ell,3+\ell} \quad (j = 1, \dots, \ell-1)$$
$$X_{2+\ell,\ell} = -X_{1+\ell,3+\ell}$$

(+ℓ) for n−1

---

we can get +0, +1, +2, ..., +(n−4), +(n−3) at the n−1 level by projecting. we only

goes from +0

goes from (−3)

$\underbrace{+1, +2, \dots, +(n-4)}_{\text{Goes from } (+\ell)}$

(−1 for n−1)

missing +(n−2) shift (downstairs) $\longrightarrow$ missing

$$\begin{cases} X_{1,j} = X_{n-1,j} - X_{n-1,2} & j = 3, \dots, n-2 \\ X_{1,n-2} = -X_{n-1,2} \end{cases}$$

57	51	52	53	54
	61	62	63	64
		72	73	74
			13	14
				24

62	63	64	—	6 n−1
	13	14	—	1 n−1
		24	—	2 n−1
				n−3, n−1

---

when projecting 2-zeros from n=7 we are missing

51	52	53	54
	62	63	64
		13	14
			24

and

62	63	64	65
	13	14	15
		24	25
			35

---

let's look at a 3-zero for n = 7

$$X_{46} = X_{16} - X_{15}$$
$$X_{46} = X_{26} - X_{25}$$
$$X_{46} = X_{36} - X_{35}$$
$$X_{47} = \quad - X_{15}$$
$$X_{47} = X_{27} - X_{25}$$
$$X_{47} = X_{37} - X_{35}$$

13	14	15	16	17
	24	25	26	27
		35	36	37
			46	47
				57

Conjecture: going from 7 → 6 we recover all 1-zero

going from 8 → 7 we recover all 1-zeros and 2-zeros

going from 9 → 8 we recover all 1-zeros and 2-zeros going from 1n → 9 we recover all 1-zeros and 2-zeros and 3-zeros

---

let's start by showing we recover all the 1-zero. We get $+0, \dots, +n-3$ from the 1-zeros, as shown.

To get the $+n-2$ one, let's start from the $+n-2$ 2-zero for n

$X_{35} = X_{15} - X_{14}$

$X_{35} = X_{25} - X_{24}$

$\left.\begin{array}{l} X_{3j} = X_{1j} - X_{14} \\ X_{3j} = X_{2j} - X_{24} \end{array}\right\}$ $j = 6, \dots, n-2$

$X_{3n-1} = X_{1n-1} - X_{14}$

$X_{3n-1} = X_{2n-1} - X_{2}$

$\boxed{13}$	14	15	16	17	— $\boxed{\ }$	$\boxed{1n}$
	$\boxed{24}$	25	26	27	— —	$\boxed{2n}$
		$\boxed{35}$	36	37	— — —	3n
			40	47	—	4n
				57		
					—	n-2,n

4 cases

Conjecture

- shift so 1 n-1 is in the bottom BF → 2-zero
- shift so 1 n-1 is in the top BF → 1-zero
- shift so 1 n-1 gets out the 24 or 35 spots → 1-zero
- shift so 1 n-1 gets in the n

13 14 15 16 17

13	14	15	~~16~~	~~17~~	2 zero
	24	25	26	~~27~~	
		35	36	~~37~~	
			46	~~47~~	
				~~57~~	

24	25	26	~~27~~	21	2 zero
	35	36	~~37~~	31	
		46	~~47~~	41	
			~~57~~	51	
				~~61~~	

35	36	~~37~~	31	32	2-zero
	46	~~47~~	41	42	
		~~57~~	51	52	
			~~61~~	62	
				~~71~~	

46	~~47~~	41	42	43	2 zero
	~~57~~	51	52	53	
		61	62	63	
			~~71~~	~~73~~	
				13	

~~54~~ 51 52 53 54   1-zero
61 62 63 64
~~71~~ 73 74
13 14
24

~~61~~ 62 63 64 65   1-zer
~~71~~ ~~73~~ ~~74~~ ~~75~~
13 14 15
24 25
35

$\boxed{-2}$

~~72 73 74 75 76~~   1-zero
13 14 15 ~~16~~
24 25 26
35 36
46

---

look at the (+n-2) 2-zero

~~n-1,1~~   n-1 2   n-1,3   n-1,4 --- n-1, n-2
~~n 2   n 3   n 4   n, n-2~~
1 3   1 4   1, n-1
2 4   2, n-2
n-4, n-2

→ this is the (+n-2) 1-zero

→ we get all the 1-zeros

---

expectation: the +n-2, +n-3, +n-4 3-zeros for n=8 project to
the +n-2, +n-3, +n-4 2-zeros for n=7

---

look at this n=7 3-zero

~~57~~ 51 52 53 54
~~61~~ 62 63 64
~~72 73 74~~
13 14
24

-3   3-zero

$\cancel{X_{13} = X_{73} - X_{72}}$
$X_{13} = X_{03} - X_{02}$
$X_{13} = X_{53} - X_{52}$
$\cancel{X_{14} = X_{74} - X_{72}}$
$X_{14} = X_{64} - X_{62}$
$X_{14} = \quad - X_{52}$

$X_{16} = 0$ →

$$\boxed{\begin{aligned} X_{13} &= X_{03} - X_{02} \\ X_{13} &= X_{53} - X_{52} \\ X_{14} &= X_{64} - X_{62} \\ X_{14} &= \quad - X_{52} \end{aligned}}$$

→ we can linearly get all the zeros

$$X_{46} = X_{16} - X_{15}$$
$$X_{46} = X_{26} - X_{25}$$
$$X_{46} = X_{36} - X_{35}$$
$$X_{47} = \quad - X_{15}$$
$$X_{47} = X_{27} - X_{25}$$
$$X_{47} = X_{37} - X_{35}$$

$$\begin{array}{ccccc} 13 & 14 & 15 & 16 & 17 \\ & 24 & 25 & 26 & 27 \\ & & 35 & 36 & 37 \\ & & & 46 & 47 \\ & & & & 57 \end{array}$$

$$X_{15} = \quad - X_{47}$$

$$X_{46} = X_{16} + X_{47}$$
$$X_{46} = X_{26} - X_{25}$$
$$X_{46} = X_{36} - X_{35}$$
$$X_{15} = \quad - X_{47}$$
$$X_{47} = X_{27} - X_{25}$$
$$X_{47} = X_{37} - X_{35}$$

$$\begin{array}{ccccc} 13 & 14 & 15 & 16 & 17 \\ & 24 & 25 & 26 & 27 \\ & & 35 & 36 & 37 \\ & & & 46 & 47 \\ & & & & 57 \end{array}$$

$$\begin{array}{ccccc} 46 & 47 & 41 & 42 & 43 \\ & 57 & 51 & 52 & 53 \\ & & 61 & 62 & 63 \\ & & & 72 & 73 \\ & & & & 13 \end{array}$$

n=5

$$\begin{array}{ccc} 13 & 14 & 15 \\ & 24 & 25 \\ & & 35 \end{array} \to \begin{cases} X_{35} = X_{25} - X_{24} \\ X_{35} = \quad - X_{14} \end{cases} \to \begin{array}{l} -X_{14} = X_{25} - X_{24} \\ X_{14} = -X_{35} \end{array} \to \boxed{\begin{array}{l} X_{25} = X_{24} - X_{14} \\ X_{14} = -X_{35} \end{array}}$$

can we shift $X_{14} = -X_{35}$ into $X_{25} = -X_{13}$?

+1 $\quad 25 = -41$

+2 $\quad 31 = -52$

$$\begin{array}{l} X_{25} = X_{24} - X_{14} \\ X_{14} = -X_{35} \end{array} \xrightarrow{+2} \begin{cases} X_{42} = X_{41} - X_{31} \\ X_{31} = -X_{52} \end{cases}$$

$$\begin{array}{l} X_{35} = X_{25} - X_{24} \\ X_{35} = \quad - X_{14} \end{array} \xrightarrow{+2} \begin{cases} X_{52} = X_{42} - X_{41} \\ X_{52} = -X_{31} \end{cases}$$

n=6

$$\begin{array}{cccc} 51 & 52 & 53 & 54 \\ & 62 & 63 & 64 \\ & & 13 & 14 \\ & & & 24 \end{array}$$

$\updownarrow$ equivalent

$$\begin{array}{cccc} 35 & 36 & 31 & 32 \\ & 46 & 41 & 42 \\ & & 51 & 52 \\ & & & 62 \end{array}$$

special:

$$\begin{array}{l} X_{24} = -X_{53} \\ X_{24} = X_{64} - X_{63} \\ X_{24} = X_{14} - X_{13} \end{array} \Leftrightarrow \begin{array}{l} X_{24} = -X_{53} \\ -X_{53} = X_{64} - X_{63} \\ -X_{53} = X_{14} - X_{13} \end{array} \Leftrightarrow \begin{array}{l} X_{46} = X_{36} - X_{35} \\ X_{41} = X_{31} - X_{35} \\ X_{42} = -X_{35} \end{array}$$

+2

---

start with +4 1-zero

$$\begin{array}{cccc} 51 & 52 & 53 & \boxed{54} \\ & 62 & 63 & 64 \\ & & 13 & 14 \\ & & & 24 \end{array} \rightarrow \begin{cases} X_{62} = X_{52} - X_{51} \\ X_{63} = X_{53} - X_{51} \\ X_{64} = \quad\; - X_{51} \end{cases}$$

equivalent to

$$\begin{array}{cccc} 13 & 14 & 15 & \boxed{16} \\ & 24 & 25 & 26 \\ & & 35 & 36 \\ & & & 40 \end{array} \quad \text{3-zero} \rightarrow \begin{cases} X_{46} = X_{36} - X_{35} \\ X_{46} = X_{26} - X_{25} \\ X_{46} = \quad\; - X_{15} \end{cases}$$

projection:

$$\begin{array}{cccc} \cancel{51} & 52 & 53 & \boxed{54} \\ & \cancel{62} & \cancel{63} & \cancel{64} \\ & & 13 & 14 \\ & & & 24 \end{array} \rightarrow \begin{array}{ccc} 52 & 53 & 54 \\ & 13 & 14 \\ & & 24 \end{array}$$

no constraint

$$\begin{array}{cccc} 13 & 14 & \cancel{15} & \cancel{\boxed{16}} \\ & 24 & 25 & \cancel{26} \\ & & 35 & \cancel{36} \\ & & & \cancel{40} \end{array} \rightarrow \begin{array}{ccc} 13 & 14 & \cancel{15} \\ & 24 & 25 \\ & & 35 \end{array}$$

no constant

$$\begin{array}{ccccc} 57 & 51 & 52 & 53 & 54 \\ & 61 & 62 & 63 & 64 \\ & & 72 & 73 & 74 \\ & & & 13 & 14 \\ & & & & 24 \end{array} \xrightarrow{\text{proj}} \begin{array}{cccc} 51 & 52 & 53 & 54 \\ & 62 & 63 & 64 \\ & & 13 & 14 \\ & & & 24 \end{array}$$

$$\begin{array}{ccccc} 57 & 51 & 52 & 53 & 54 \\ & 61 & 62 & 63 & 64 \\ & & 72 & 73 & 74 \\ & & & 13 & 14 \\ & & & & 24 \end{array} \xrightarrow{\text{proj}} \begin{array}{cccc} 51 & 52 & 53 & 54 \\ & 62 & 63 & 64 \\ & & 13 & 14 \\ & & & 24 \end{array}$$

Confusion:

- 4 of the 2-zeros project to 2-zero
- 3 of the 2-zeros project to 1-zeros

Somehow we recover the remaining 2-zeros (2 of them) from the 3-zeros, but they are equivalent to 2-zeros

---

13 14 15 16 17
24 25 26 27
35 36 37
40 47
57
→
+0
13 14 15 16
24 25 26
35 36
40

13 14 15 16 17
24 25 26 27
35 36 37
40 47
57
→
13 14 15 16
24 25 26
35 36
40

24 25 26 27 21
35 36 37 31
40 47 41
57 51
61
→
+1
24 25 26 21
35 36 31
40 41
51

24 25 26 27 21
35 36 37 31
40 47 41
57 51
61
→
24 25 26 21
35 36 31
40 41
51

35 36 37 31 32
40 47 41 42
57 51 52
61 62
72
→
+2
35 36 31 32
40 41 42
51 52
62

35 36 37 31 32
40 47 41 42
57 51 52
61 62
72
→
35 36 31 32
40 41 42
51 52
62

40 47 41 42 43
57 51 52 53
61 62 63
72 73
13
→
+3
40 41 42 43
51 52 53
62 63
13

40 47 41 42 43
57 51 52 53
61 62 63
72 73
13
→
40 41 42 43
51 52 53
62 63
13

57 51 52 53 54
61 62 63 64
72 73 74
13 14
24
→
51 52 53 54 +4
62 63 64
13 14
24

57 51 52 53 54
61 62 63 64
72 73 74
13 14
24
→
+4
51 52 53 54
62 63 64
13 14
24

61 62 63 64 65
72 73 74 75
13 14 15
24 25
35
→
62 63 64 65
13 14 15 +5
24 25
35

61 62 63 64 65
72 73 74 75
13 14 15
24 25
35
→
+5
62 63 64 65
13 14 15
24 25
35

72 73 74 75 76
13 14 15 16
24 25 26
35 36
46
→
13 14 15 16 +0
24 25 26
35 36
46

72 73 74 75 76
13 14 15 16
24 25 26
35 36
46
→
+6
13 14 15 16
24 25 26
35 36
46

13 14 15 16 17
24 25 26 27
35 36 37
40 47
57

→

13 14 15 16
24 25 26
35 36
40

+0

24 25 26 27 21
35 36 37 31
40 47 41
57 51
61

→

24 25 26 21
35 36 31
40 41
51

+1

35 36 37 31 32
40 47 41 42
57 51 52
61 62
72

→

35 36 31 32
40 41 42
51 52
62

+2

40 47 41 42 43
57 51 52 53
61 62 63
72 73
13

→

40 41 42 43
51 52 53
62 63
13

+3

57 51 52 53 54
61 62 63 64
72 73 74
13 14
24

→

51 52 53 54
62 63 64
13 14
24

+4

61 62 63 64 65
72 73 74 75
13 14 15
24 25
35

→

62 63 64 65
13 14 15
24 25
35

X

72 73 74 75 76
13 14 15 16
24 25 26
35 36
46

→

13 14 15 16
24 25 26
35 36
46

X

---

current conjecture: we can get everything

$$\begin{array}{ccccc} 24 & 25 & 26 & 27 & 21 \\ & 35 & 36 & 37 & 31 \\ & & 40 & 47 & 41 \\ & & & 57 & 51 \\ & & & & 61 \end{array} \rightarrow \begin{array}{cccc} \overset{+1}{} & & & \\ 24 & 25 & 26 & 21 \\ & 35 & 36 & 31 \\ & & 40 & 41 \\ & & & 51 \end{array} \rightarrow \begin{cases} X_{46} = X_{36} - X_{35} \\ X_{46} = X_{26} - X_{25} \\ X_{41} = X_{31} - X_{35} \\ X_{41} = \quad\; - X_{25} \end{cases}$$

$\updownarrow$ equivalent $\qquad\qquad$ $\updownarrow$ equivalent

$$\begin{array}{cccccc} 57 & 51 & 52 & 53 & 54 \\ & 61 & 62 & 63 & 64 \\ & & 72 & 73 & 74 \\ & & & 13 & 14 \\ & & & & 24 \end{array} \rightarrow \begin{array}{cccc} \overset{+4}{} & & & \\ 51 & 52 & 53 & 54 \\ & 62 & 63 & 64 \\ & & 13 & 14 \\ & & & 24 \end{array} \rightarrow \begin{cases} X_{13} = X_{63} - X_{62} \\ X_{13} = X_{53} - X_{52} \\ X_{14} = X_{64} - X_{62} \\ X_{14} = \quad\; - X_{52} \end{cases}$$

$$\begin{aligned} X_{13} &= X_{63} - X_{62} \\ X_{13} &= X_{53} - X_{52} \\ X_{46} &= X_{62} - X_{25} \\ X_{14} &= \quad\; - X_{52} \end{aligned} \rightarrow \begin{aligned} X_{13} &= X_{63} - X_{46} + X_{14} \\ X_{13} &= X_{53} + X_{14} \\ X_{46} &= X_{62} - X_{25} \\ X_{14} &= \quad\; - X_{52} \end{aligned} \rightarrow \left| \begin{aligned} X_{46} &= X_{63} - X_{53} \\ X_{14} &= X_{13} - X_{35} \\ X_{46} &= X_{62} - X_{25} \\ X_{14} &= \quad\; - X_{52} \end{aligned} \right.$$

↳ not all 2-zeros are independent! $(n=7)$

Focus going forward: prove there are no crossings with a short.